

Autodesk® Scaleform®

Scaleform 4.2

Scaleform 4.2

Michael Antonov

1.03

2012 8 2

Copyright Notice

Autodesk® Scaleform® 4.2

© 2012 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AUGI, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo) Built with ObjectARX (design/logo), Burn, Buzzsaw, CAiCE, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, DesignServer, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWFX, DXF, Ecotect, Evolver, Exposure, Extending the Design Team, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, Freewheel, GDX Driver, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, RiverCAD, Robot, Scaleform, Scaleform GfX, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, Tinkerbox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

How to Contact Autodesk Scaleform:

Document	Scaleform 4.2 Renderer Threading Guide
Address	Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

1		1
2		1
2.1		1
2.2		2
2.3		3
2.4		4
2.5		5
3		6
3.1		6
4		8
5		10
5.1		10
5.2		10
5.3		12
6	GfxShaderMaker	13
6.1		13
6.2	API HAL.....	13
6.2.1	D3D9	13
6.2.2	D3D1x	13

Scaleform 4.0 (Advance)

Scaleform SDK

Scaleform 4.0 API

Scaleform

2

API	Scaleform3.x	Scaleform4
-----	--------------	------------

1. Renderer2D

2. GfxLoader

3.

4. MovieDisplayHandle

Renderer2D::Display

5. Gfx::Movie

6. API

GRenderer Scaleform 3.X Render::HAL

2.1

Scaleform	Render::Renderer2D	Render::HAL	Renderer2D
	Display	Render::HAL	Renderer2D
“ ”		Render::D3D9::HAL	

Render::PS3::HAL

```
D3DPRESENT_PARAMETERS    PresentParams;
IDirect3DDevice9*         pDevice = ...;
ThreadId                  renderThreadId = Scaleform::GetCurrentThreadId();
Render::ThreadCommandQueue *commandQueue = ...;
Ptr<Render::D3D9::HAL>    pHal;
Ptr<Render::Renderer2D>  pRenderer;

pHal = *SF_NEW Render::D3D9::HAL(commandQueue);
if (!pHal-> InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
```



```

GlyphCacheParams gcp;
gcp.TextureWidth  = 1024;
gcp.TextureHeight = 1024;
gcp.NumTextures   = 1;
gcp.MaxSlotHeight = 48;

pRenderer->GetGlyphCacheConfig()->SetParams(gcp);

```

Scaleform 3.3 TextureWidth TextureHeight NumTextures
MaxSlotHeight

InitHAL InitHAL
SetParams

2.3

Scaleform 4.0 GFx::ImageCreator Render::TextureManager
Scaleform ImageCreator
GFx::Loader

```

GFx::Loader loader;
...

SF::Ptr<GFx::ImageCreator> pimageCreator =
    *new GFx::ImageCreator(pRenderThread->GetTextureManager());
loader.SetImageCreator(pimageCreator);

```

ImageCreator TextureManager TextureManager
Render::HAL::GetTextureManager()
GetTextureManager

Render::HAL TextureManager CreateTexture

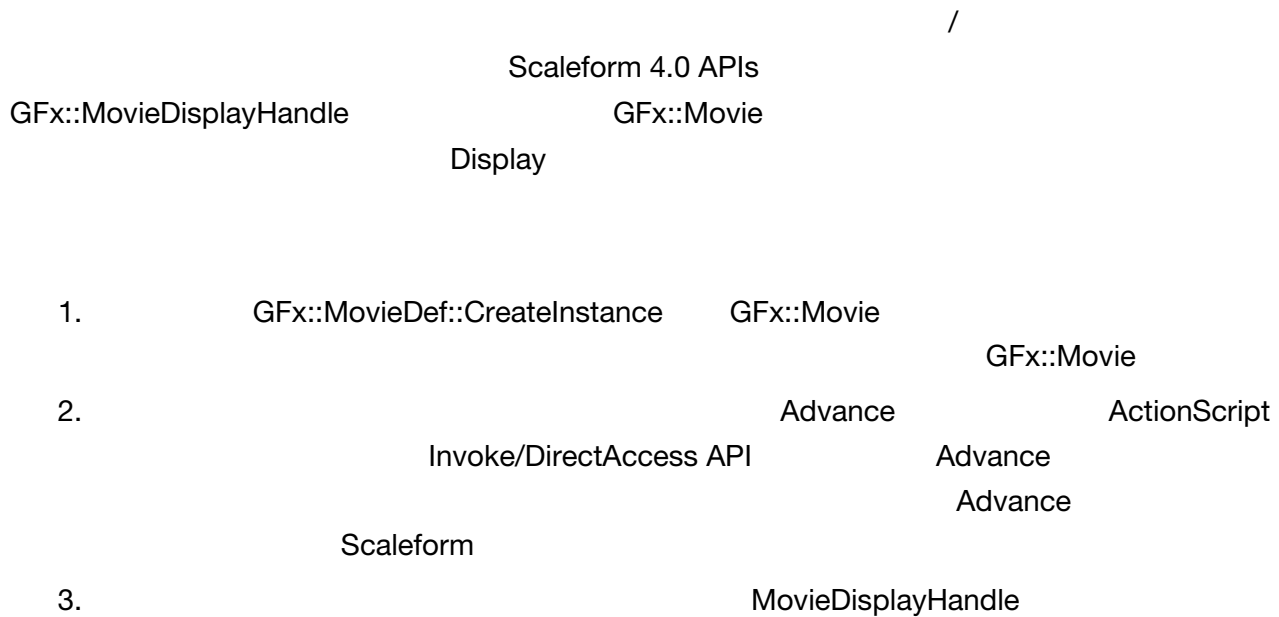
D3DCREATE_MULTITHREADED

D3D9

- D3DCREATE_MULTITHREADED D3D9
- ThreadCommandQueue Render::HAL
- D3D ThreadCommandQueue

Platform::RenderHALThread

2.4



```
//-----
// 1.
Ptr<GfX::Movie> pMovie = ...;
GfX::MovieDisplayHandle hMovieDisplay;

//
//
pMovie->SetViewport(width, height, 0,0, width, height);
hMovieDisplay = pMovie->GetDisplayHandle();

// Scaleform Player
//
pRenderThread->AddDisplayHandle(hMovieDisplay);
```



```

//-----
// 2.

//
//   ExternalInterface      Advance
//   float deltaT = ...;
//   pMovie->HandleEvent(...);
//   pMovie->Advance(deltaT);

//
//
//   pRenderThread->WaitForOutstandingDrawFrame();
//   pRenderThread->DrawFrame();

//-----
// 3.      -      DrawFrame
//   Ptr<Render::Renderer2D> pRenderer = ...;

//   pRenderer->BeginFrame();
//   bool hasData = hMovieDisplay.NextCapture(pRenderer->GetContextNotify());
//   if (hasData)
//       pRenderer->Display(hMovieDisplay);
//   pRenderer->EndFrame();

```

- WaitForOutstandingFrame – CPU
- Advance
- MovieDisplay.NextCapture: “ ”Advance false

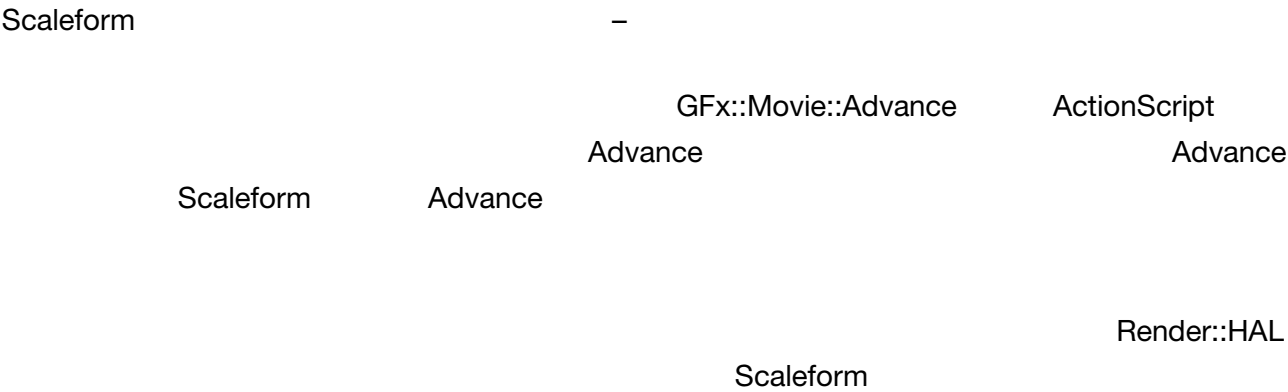
2.5

Gfx::Movie Release

Render::ThreadCommandQueue Movie
ThreadCommandQueue

Movie::ShutdownRendering(false)
Movie::IsShutdownRenderingComplete
NextCapture false

3



•

•

Scaleform
Player Scaleform

3.1

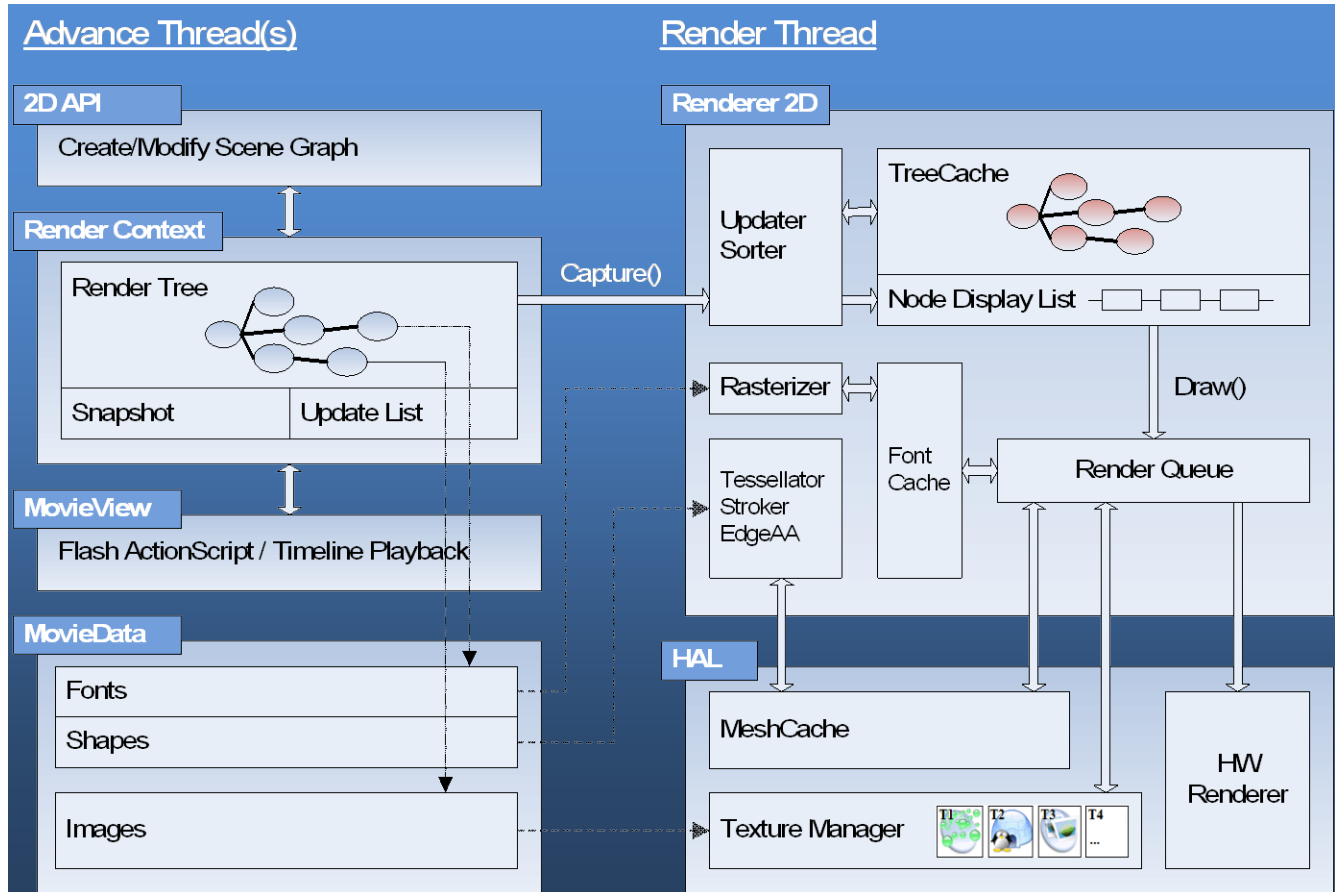
Advance
/

Scaleform
Gfx::Movie::Advance
Gfx::Movie::Capture

NextCapture

- Movie::Advance
- Capture NextCapture Capture Capture
- Capture Advance Invoke Direct Access API
Movie
- Scaleform 4.0 Scaleform 3.3
Display Movie::Invoke
- Advance Capture Advance Invoke
Advance

Scaleform 4.0



1

Scaleform

Movie

Render Context

2D

Movie

MovieDef

Render Tree

Scaleform

Render::Renderer2D

Display

Render::HAL

Renderer2D

Render::HAL

Render HAL

- Font Cache
 - Mesh Cache
 - Render Queue / Render Queue
 - HAL API
- JavaScript / ActionScript
- CPU

5

Scaleform

5.1

```

    ,
    rendererString    rendererFloat    ActionScript    ,
    setRendererFloat  AS2              AS3              setRendererString
    AS2/AS3
    ,
    (Float),          HAL::PushUserData    UserDataState::Data
Render::HAL    UserDataStack
    ,
    ,
    UserDataState::Data    HAL::PopUserData
    /
    /
```

5.2

```

    ,
    HAL
    ,
    HAL
    ,
    2D
    AS3
    SWF,
    m
    AS3
```

```
import scaleform.gfx.*;
DisplayObjectEx.setRendererString(m, "Abc");
DisplayObjectEx.setRendererFloat(m, 17.1717);
```

```
4.1
Src/Render/ShaderData.xml ,
```

```
<ShaderFeature id="Position">
  <ShaderFeatureFlavor id="Position2d" hide="true"/>
  <ShaderFeatureFlavor id="Position3d"/>
  <ShaderFeatureFlavor id="Position2dOffset"/>
</ShaderFeature>
```



```

if (pHal->UserDataStack.GetSize() > 0 )
{
    const UserDataState::Data* data = pHal->UserDataStack.Back();
    if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
        data->RendererString.CompareNoCase("abc") == 0)
    {
//    poffset
        for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
        {
            float poffset = data->RendererFloat / 256.0f;
            SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
        }
    }
},
    'meshCount'
    (
    1    HAL    ,    'm'
    ),meshCount

```

5.3

```

HAL    ,    Scaleform
    ,
    ,    Scaleform
    Scaleform
    /
disableBatching    AS3    disableBatching    ,    AS2
    AS2/AS3
    ,

```


6 GFxShaderMaker

6.1

Scaleform 4.1 GFxShaderMaker, Scaleform
(Render_D3D9) ,
ShaderData.xml
, /
, D3D9 ,

Src/Render/D3D9_ShaderDescs.h
Src/Render/D3D9_ShaderDescs.cpp
Src/Render/D3D9_ShaderBinary.cpp

, ,

6.2 *API HAL*

6.2.1 D3D9

D3D9 HAL ShaderModel 2.0 ShaderModel 3.0 ,HAL

GFxShaderMaker '-shadermodel'

GFxShaderMaker.exe -platform D3D1x -shadermodel SM30

ShaderModel 2.0 ShaderModel 2.0
InitHAL,

6.2.2 D3D1x

D3D1x HAL ()
D3D11 D3D10.1
FEATURE_LEVEL_10_0
FEATURE_LEVEL_9_3
FEATURE_LEVEL_9_1

,HAL

GFxShaderMaker '-featurelevel'

GFxShaderMaker.exe -platform D3D1x -featurelevel
FEATURE_LEVEL_10_0,FEATURE_LEVEL_9_1

FEATURE_LEVEL9_3
InitHAL,

(
FEATURE_LEVEL_9_3
FEATURE_LEVEL_9_1)